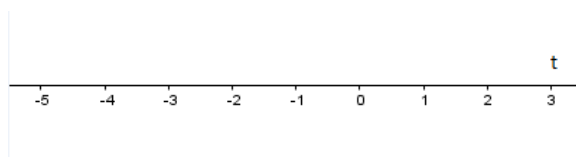


Problem 1

Problem 1

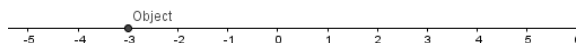
Problem 1 The position, $s(t)$, of an object moving along a horizontal line is given by $s(t) = t^2 - 4$, where s is in meters and t is in seconds, $0 \leq t < 5$.

- (a) Mark the position of the object on the line at time $t = 1$:



Explanation. $s(1) = 1^2 - 4 = -3$

Position of object on line
at time $t = 1$.



- (b) Find the average velocity, v_{AV} , of the object during the time interval $[1, 3]$.

Explanation. The average velocity over $[1, 3]$ is

$$\frac{s(3) - s(1)}{3 - 1} = \frac{5 - (-3)}{2} = \frac{8}{2} = 4 \text{ m/s}$$

- (c) Compute the average velocity, $v_{AV}(t)$, of the object during the time interval

- (i) $[1, t]$, for $1 < t < 5$;

Explanation. The average velocity, $v_{AV}(t)$ over $[1, t]$ is

$$\begin{aligned} v_{AV}(t) &= \frac{s(t) - s(1)}{t - 1} = \frac{(t^2 - 4) - (-3)}{t - 1} \\ &= \frac{t^2 - 1}{t - 1} = t + 1, \quad 1 < t < 5 \end{aligned}$$

- (ii) $[t, 1]$, for $0 \leq t < 1$.

Explanation. The average velocity over $[t, 1]$ is

$$\begin{aligned} v_{\text{AV}}(t) &= \frac{s(1) - s(t)}{1 - t} = \frac{(-3) - (t^2 - 4)}{1 - t} \\ &= \frac{1 - t^2}{1 - t} = 1 + t, \quad 0 < t < 1 \end{aligned}$$

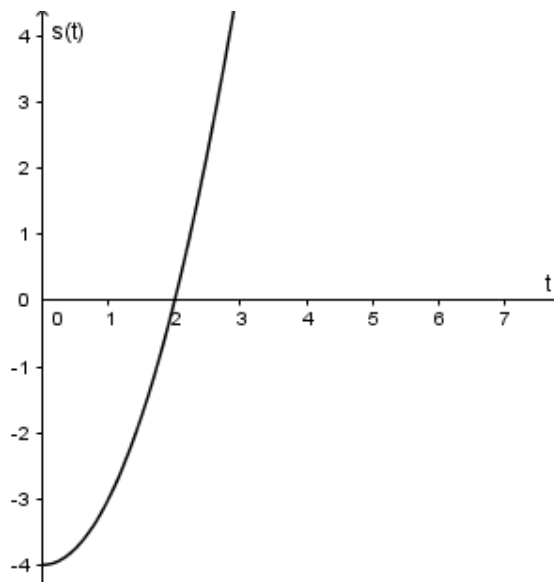
$$\text{Note: } v_{\text{AV}}(t) = \frac{s(1) - s(t)}{1 - t} = \frac{s(t) - s(1)}{t - 1} = 1 + t, \quad 0 \leq t < 5$$

- (d) Find the instantaneous velocity, v_{inst} , of the object at $t = 1$. Justify your answer.

Explanation. The instantaneous velocity of the object at $t = 1$ is given by $\lim_{t \rightarrow 1} \frac{s(t) - s(1)}{t - 1}$. This limit has form $\frac{0}{0}$.

$$\begin{aligned} v_{\text{inst}} &= \lim_{t \rightarrow 1} v_{\text{AV}}(t) = \lim_{t \rightarrow 1} \frac{s(t) - s(1)}{t - 1} \\ &= \lim_{t \rightarrow 1} (t + 1) = 2. \end{aligned}$$

- (e) The position-time graph of the function s is given in the figure below.



- (i) Assume P is a point on the graph of s . Fill in the blank.

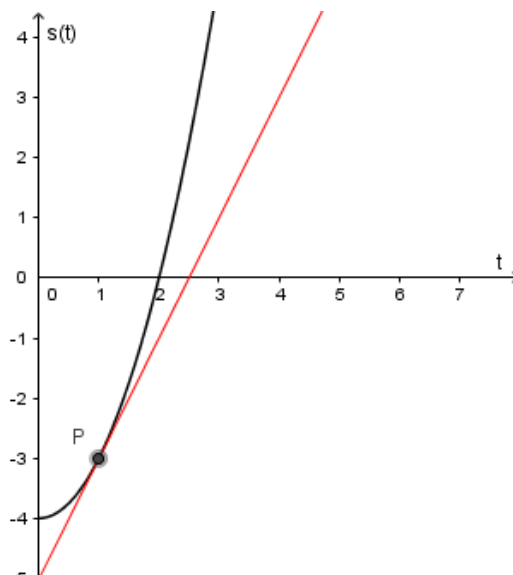
$$P = (1, \underline{\quad}).$$

Explanation. $P = (1, \underline{-3})$

Problem 1

- (ii) Plot the point P and draw the tangent line at this point in the figure above.

Explanation.



- (iii) Find the slope, m_{tan} , of the tangent line in part (ii). Explain.

Explanation. The slope of the tangent line at $t = 1$ is the same as the instantaneous velocity at $t = 1$. Therefore $m_{\text{tan}} = v_{\text{inst}} = 2$.